BENJAMIN DAVIES

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Education

Stanford University
PhD in Economics 2020–
Fields: Microeconomic Theory, Behavioral and Experimental Economics
Committee: Matthew Jackson (co-advisor), Arun Chandrasekhar (co-advisor), Steven Callander

University of Canterbury
BSc(Hons, 1st class) in Economics and Mathematics 2014–17

Working Papers

Course design in the age of AI

Abstract: I develop a model of learning-by-doing and course design, and use it to study the impacts of artificial intelligence (AI). A myopic student faces a sequence of tasks that he can work on or delegate to AI. Work requires costly effort but builds skill; delegation requires no effort but builds no skill. A teacher designs the task sequence ("course") to maximize the student's skill development, given his choices to work or delegate. Without AI, the teacher makes earlier tasks more effort-intensive and later tasks more skill-intensive. With AI, the teacher must redesign early tasks to induce effort, leading to less skill development. If AI complements effort, then improvements in AI quality make high-skill students learn faster but low-skill students learn slower.

The value of conceptual knowledge (with Anirudh Sankar)

Abstract: We define and quantify the instrumental value of conceptual knowledge. Such knowledge tells agents how unknown, payoff-relevant states relate. It is distinct from the statistical knowledge gained from data on those states. We formalize this distinction in a Bayesian decision framework with Gaussian priors and quadratic losses. Conceptual knowledge is valuable because it empowers agents to collect more valuable data. It is more valuable when states are more "reducible": when they can be explained with fewer common concepts. Its value is non-monotone in sample size and vanishes when samples have infinite size. Agents who know more concepts can attain the same payoffs with less data. This is especially true when states are highly reducible.

How mechanistic explanations reshape learning and behavior: Evidence from a fertilizer choice experiment in Eastern Uganda (with Anirudh Sankar, Robert Dulin, Vesall Nourani, Jess Rudder, Abraham Salomon, and Godfrey Taulya)

Abstract: Mechanistic explanations play a key role in human cognition and scientific progress. Despite their importance, we lack systematic evidence on whether and how mechanistic explanations help lay decision-makers interpret information in complex economic environments. We evaluate the causal impact of including mechanistic explanations in an information intervention: public demonstrations of fertilizer use for smallholder tomato farmers in Eastern Uganda. In all demonstrations, extension officers showcased the impact of a recommended fertilizer recipe. In the treatment group, officers also explained the mechanisms underlying the recipe's effects—introducing the language of macronutrients and the causal processes linking nutrients, soil features, and plant growth. We collected detailed data on beliefs and behaviors from 797 farmers in a lab-in-the-field experiment conducted at the demonstration site and followed up with them over two growing seasons. In the lab-in-the-field, treated farmers generalized more effectively and achieved 9% higher simulated profits in an incentivized fertilizer application task. At endline, treated farmers' real fertilizer choices reflected improved nutrient timing and balance, and their yields were 14% higher.

Peer-Reviewed Publications

Persistence, patience and costly information acquisition
Economics Letters, 2026

Abstract: A forward-looking agent observes signals of a state that follows a Gaussian AR(1) process. He balances the cost of having imprecise beliefs with the cost of acquiring more precise signals. I characterize his optimal information acquisition policy, and analyze how his steady-state beliefs and costs depend on persistence (the AR(1) parameter) and patience (the agent's discount factor). Higher persistence has a non-monotone effect on belief precision and raises overall costs. Higher patience makes beliefs more precise and lowers overall costs.

Gender sorting among economists: Evidence from the NBER
Economics Letters, 2022

Abstract: I compare the co-authorship patterns of male and female economists, using historical data on National Bureau of Economic Research working papers. Men tended to work in smaller teams than women, but co-authored more papers and so had more co-authors overall. Both men and women had more same-gender co-authors than we would expect if co-authorships were random. This was especially true for men in Macro/Finance.

COVID-19, lockdown and two-sided uncertainty (with Arthur Grimes)
New Zealand Economic Papers, 2022

Abstract: When COVID-19 struck, the government had two choices: enter lockdown immediately or delay its decision. Delaying would have allowed more information to emerge about health and economic impacts, and preserved the option to act later. However, delaying may also have destroyed the option to eradicate COVID-19. We model the government's decisions under the health and economic uncertainty generated by COVID-19. Our model captures both two-sided uncertainty and the dynamic consequences that flow from the government's initial decision.

Research funding and collaboration (with Jason Gush, Shaun C. Hendy, and Adam B. Jaffe)
Research Policy, 2022

Abstract: We analyze whether research funding contests promote co-authorship. Our analysis combines Scopus publication records with data on the Marsden Fund, the premier source of funding for basic research in New Zealand. We use fixed-effect models to analyze within-researcher-pair variation in co-authorship. Among pairs who ever co-authored or co-proposed, co-authorship was 13.8 percentage points more likely in a given year if they had co-proposed during the previous ten years than if they had not. This co-authorship rate was not significantly higher among funded pairs. However, when we increase post-proposal publication lags towards the length of a typical award, we find that funding, rather than participation, promotes co-authorship.

Relatedness, complexity and local growth (with David C. Maré)
Regional Studies, 2021

Abstract: We derive a measure of the relatedness between economic activities based on weighted correlations of local employment shares. Our approach recognizes variation in the extent of local specialization and adjusts for differences in data quality between cities. We use our measure to estimate activity and city complexity, and examine the contribution of relatedness and complexity to urban employment growth in New Zealand. Relatedness and complexity are complementary in promoting employment growth in New Zealand's largest cities, but do not contribute to employment growth in its smaller cities.

Awards

William Georgetti Scholarship (for "best brains" in New Zealand)	2020
Seamus Hogan Memorial Prize (for best thesis in economics at Canterbury)	2017
Cook Memorial Prize (for top honors mathematics student at Canterbury)	2017
Sir Frank Holmes Prize (for top undergraduate economics student in New Zealand)	2016

Research and Professional Experience

Stanford University
Research Assistant to Matthew Jackson 2021–22

Motu Economic and Public Policy Research
Research Analyst 2018–20

PwC New Zealand
Actuarial Consulting Intern 2015–16

Teaching Experience

Stanford University
Teaching Assistant for ECON 198 (Junior Honors Seminar) 2026
Teaching Assistant for ECON 178 (Behavioral Economics) 2025
Teaching Assistant for ECON 137 (Decision Modeling and Information) 2025
Teaching Assistant for BUSGEN 143 (Finance, Corporations, and Society) 2025
Teaching Assistant for ECON 43 (Personal Finance) 2023–24
Teaching Assistant for ECON 135 (Foundations of Finance) 2022–24

University of Canterbury
Teaching Assistant for FINC 331 (Financial Economics) 2017
Teaching Assistant for ECON 208 (Intermediate Microeconomics) 2016–17
Teaching Assistant for ECON 321 (Microeconomic Theory) 2016–17

Conference Presentations

Society for Judgment and Decision Making Annual Conference (Poster), San Francisco 2023
NBER Summer Institute (Science of Science Funding) 2021
American Economic Association Annual Meeting 2021
New Zealand Association of Economists Conference, Wellington 2019
New Zealand Association of Economists Conference, Auckland 2018
European Group of Risk and Insurance Economists Annual Seminar, London 2017